

Lewy

Protein AI for Parkinson's disease

ESM-2, the world's leading protein AI, predicts Parkinson's variants backwards. We fix that.

Evan O'Leary + Niall O'Leary (patient co-founder)

THE PROBLEM

Current protein AI fails on Parkinson's - silently.

\$4.9B/yr

invested in PD drug discovery annually

-0.17

Spearman rho - ESM-2 on SNCA (vs +0.41 on normal proteins)

Backwards

ESM-2 ranks protective variants as damaging, and vice versa

*"When a clinician sees an unknown SNCA variant, they run ESM-2.
LewyGym shows that model has been misleading them the whole time.
The tools are broken for the most important Parkinson's gene."*

- LewyGym preprint, O'Leary & O'Leary, 2026

THE SOLUTION

A variant effect API that actually works for GOF proteins.

- 1 Fine-tune** Retrain protein LMs on aggregation-specific DMS data, not
- 2 Benchmark** Evaluate every model against LewyGym - the only PD varia
- 3 Ship API** POST /score {gene, variant} -> calibrated pathogenicity sco
- 4 Expand** SNCA first, then LRRK2, then ALS (SOD1, TDP-43), then A

ESM-2 on SNCA today

Spearman rho = -0.17 (WRONG)

Lewy API (target)

Spearman rho > +0.30 (calibrated)

We already proved the problem.

- **LewyGym benchmark** Published on GitHub + HuggingFace. 13,560 scored SNCA variants. ProteinGym-compatible format.
- **ESM-2 baseline results** 8M and 650M models benchmarked. Mean rho -0.139 / -0.171. Numbers are in the repo.
- **bioRxiv preprint** lewygym_preprint.pdf ready to submit. 5 pages, 3 tables, 9 references. Two co-authors.
- **Patient co-founder** Niall O'Leary (PD diagnosis). Not an advisory role - listed on the paper, in the loop on every decision.
- **Open source moat** CC0 data + open benchmark = community adoption. Lewy becomes the standard for PD variant evaluation.

MARKET

Every pharma company working on neurodegeneration needs this.

\$4.9B

**PD drug discovery
annual spend**

Beams, AstraZeneca, Sanofi,
Biogen - all have PD programs

\$18B

**Neurodegeneration
drug pipeline value**

ALS, Alzheimer's, Huntington's
- same GOF model failure

\$2.1B

**Computational biology
tools market**

Growing 15% YoY - pharma
paying for AI variant tools

Business model

API (per query)	\$0.01-0.10 per variant scored. Self-serve, instant.
Enterprise	\$50K-200K/yr annual license. Full gene panel, SLA, custom models.
Data licensing	Aggregation DMS datasets sold to model trainers and pharma.

TEAM + ASK

Two co-founders. One of us has Parkinson's.

Evan O'Leary

Technical co-founder

Built LewyGym and scoring harness. ESM-2 benchmarking.
Protein ML, Python, HuggingFace stack.

Niall O'Leary

Patient co-founder

Living with Parkinson's disease. Co-author on the preprint.
Provides patient perspective on every product decision.

Hiring: wet lab collaborator, ML engineer (protein models)

Pre-seed ask

\$500K

18 months runway

Use of funds:

40% Aggregation DMS experiment
(contract Arctoris / CRO)

30% Model training + API build

20% Wet lab hire

10% Ops, legal, cloud

Milestone: $\rho > +0.30$ on SNCA

= Series A ready

Lewy

Protein AI for Parkinson's disease

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github.com/Tyronita/LewyGym

huggingface.co/datasets/EvanOLeary/LewyGym